

Pulmonary Function Tests (PFTs)

INTERPRETATION PATTERNS

- **FEV1/FVC < 70% (Obstructive Defect):** The flow-volume loop shows a "scooped-out" appearance.
 - If FEV1 increases by **>12% AND >200 mL** after bronchodilator, obstruction is reversible (**Asthma**).
 - If irreversible: **COPD**.
 - DLCO is low in emphysema, normal in chronic bronchitis and asthma.
- **Restrictive pattern (TLC < 80% with normal/high FEV1/FVC):** Check DLCO.
 - Low DLCO → intrinsic (**Interstitial Lung Disease**, Sarcoidosis).
 - Normal DLCO → extrapulmonary (**Obesity, neuromuscular/chest wall**).

Asthma

- Episodic cough, wheezing, dyspnea. Worse at night, with exercise, or with triggers.

DIAGNOSTIC PATHWAY

- **Best Initial Test:** Spirometry (PFTs).
- Diagnosis confirmed if FEV1 improves by **>12% AND >200 mL** after bronchodilator.
- Normal spirometry but high suspicion: order a **methacholine challenge**. A $\geq 20\%$ drop in FEV1 supports airway hyperreactivity. High negative predictive value—if negative, asthma is unlikely.

TRAP: Vocal cord dysfunction: asthma-like symptoms despite therapy, often with inspiratory stridor or monophonic wheeze. Treat with **speech therapy**.

ASTHMA VARIANTS

- **Exercise-Induced Bronchoconstriction:** Dyspnea occurs 10-15 minutes after starting exercise. Rx: SABA 15 minutes before exercise.
- **Aspirin-Exacerbated Respiratory Disease (AERD):** Asthma + Chronic Rhinosinusitis with Nasal Polyps + Reaction to Aspirin/NSAIDs (**Samter's Triad**).
- **Allergic Bronchopulmonary Aspergillosis (ABPA):** Asthma + Recurrent infiltrates + Eosinophilia + High IgE (>1000). Treatment: systemic steroids (*suppress inflammatory response*) + Itraconazole (*reduces Aspergillus burden*).
- **Occupational Asthma:** Symptoms improve on weekends/vacations. Diagnosis: Peak flow monitoring at work vs home.

Core Principles for Asthma Management (GINA Updates):

1. Stepwise Therapy:

- **Preferred Reliever:** Low-dose ICS-Formoterol as needed (*reduces exacerbations; provides bronchodilation plus anti-inflammatory effect*).
- **Alternative Reliever:** SABA (Albuterol).